Numerical Computing II

Assignment-1

Assignment submitted by Khumba Lunganlung MS2515,3rd Sem 2026

1. What problem are you trying to solve?

As a researcher, my main goal is to develop an artificial intelligence system that can learn to play games through experience rather than relying on someone to manually program every possible move.

For example, when a human plays a video game, they gradually learn which actions are useful and which actions lead to failure. I want to develop an AI agent that can learn in a similar way. The agent interacts with the game, receives rewards or penalties, and improves its decisions over time.

In our research, we used Deep Q-Networks (DQN), which combine reinforcement learning with deep neural networks. The objective is to train an agent to select actions that maximize its long-term reward.

2. Why is an analytical solution difficult or impossible?

An exact mathematical solution is difficult because the number of possible game situations and action sequences can become extremely large ,averaging around 100,000 times at the least.

In some games, the agent receives only visual information, such as images of the game screen, instead of a simple numerical description of the current situation hence It is difficult to manually derive the best action for every possible screen and situation.

Therefore, instead of calculating the perfect solution beforehand, we use numerical methods and learning algorithms to estimate which actions are likely to produce better results.

3. What numerical computation is required?

Several numerical computations are required in this research.

First, the neural network processes the game-screen information and converts it into numerical representations. It then estimates the expected future reward associated with different actions.

The agent uses these estimates to select actions and compares its predictions with learning targets calculated from its experience. The difference between the prediction and the target is used to calculate an error, and the network's weights are updated using gradient-based optimization.

The main computations include:

- Matrix multiplication and other linear algebra operations.
- Neural network forward propagation and backpropagation.
- Calculation of rewards and discounted future rewards.
- Estimation of Q-values for possible actions.
- Numerical optimization to update network weights.
- Statistical evaluation of the agent's performance over multiple games.

These computations are repeated many times during training, allowing the agent to improve its policy gradually.

4. Which topics from Numerical Computing II are involved?

From the perspective of my research, several topics from Numerical Computing II are relevant.

- Linear algebra: Matrix and vector operations are essential for neural network computations.
- Numerical optimization: Gradient-based methods update the network parameters to reduce prediction errors.
- Numerical approximation: The neural network estimates action values instead of calculating exact optimal values for every possible state.
- Iterative methods: Training repeatedly updates the network until its performance improves.
- Probability and statistics: Rewards, exploration, and performance measurements involve uncertainty and statistical evaluation.

5. What software or libraries are used?

Python with PyTorch or TensorFlow, along with NumPy for numerical operations. These tools make it easier to construct neural networks, calculate gradients, update parameters, and evaluate learning performance.

6. What surprised you most about this research?

What surprised me most was that an AI agent could learn to play a variety of video games directly from screen images and reward signals, without being given a manually designed strategy for every game.

The agent learns through repeated interaction, making mistakes and adjusting its decisions based on the results.
Based on these…It's safe to say, the more advanced + efficient computing methods are used, the better these AI perform and it's also a real-life application of these knowledges.

Interviewed Mingma Tamang, IT personnel.